

■ A.2.6 Symbolic Summation

As a final example of the implementation of mathematical algorithms in *Mathematica*, we consider a somewhat more complicated example: Gosper's algorithm for finding closed-form results for symbolic sums.

This algorithm can take sums whose successive terms have ratios which are rational functions, and can find explicit formulae for sums in terms of rational and other functions. The algorithm is in some respects analogous to the Risch algorithm that *Mathematica* uses for indefinite integration.

```
Gosper::usage =
  "Gosper[term, {var, n0, n1}] finds the value of
  Sum[term, {var, n0, n1}] for symbolic n0, n1."

(* linearity *)
Gosper[an_ + bn_, {var_, n0_, n1_, inc_}] :=
  Gosper[an, {var, n0, n1, inc}] + Gosper[bn, {var, n0, n1, inc}]

(* default increment *)
Gosper[an_, {var_, n0_, n1_}] := Gosper[an, {var, n0, n1, 1}]

(* default startvalue *)
Gosper[an_, {var_, n1_}] := Gosper[an, {var, 1, n1, 1}]

(* trivial cases *)
Gosper[an_, {var_, n0_, n1_, 1}] := an (n1 - n0 + 1) /; FreeQ[an, var]
Gosper[an_, {var_, n0_, n0_, _}] := an /. var -> n0

Begin["Gosper`private`"]

Subvar[expr_, old_Symbol, new_] := expr /. old -> new

Gosper[an_, {var_, n0_, n1_, 1}] :=
Block[{pn, qn, rn, an1, rnj, jj, res, resn, resp, rat, ann, qrpos, ii,
  qrneg, dqrpos, dqrneg, dk, dp, fn, sn, k0, fn, eq, i, s0, s1},
  ann = an;
  an1 = Subvar[ann, var, var-1];
  rat = ann/an1; (* user-defined rules take effect here *)
  rat = Together[rat];
  qn = Numerator[rat];
  rn = Denominator[rat];
  If[!PolynomialQ[qn, var] || !PolynomialQ[rn, var],
    (* not a rational function *)
    Return[Sum[an, {var, n0, n1, 1}]]];
  pn = 1;
  rnj = Expand[Subvar[rn, var, var+jj]];
  res = Resultant[qn, rnj, var];
```

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```

res = FactorList[res];
res = Map[#[[1]]&, res];
(* find factors linear in jj *)
resp = Cases[res, jj+_Integer];
resn = Cases[res, -jj+_Integer];
res = Join[-(resp - jj), resn + jj];
res = Select[res, Positive];
(* adjust for positive integer zeros *)
For[resn = Length[res], resn > 0, resn--, Block[{gn, gnj},
    gn = GCD[qn, Expand[Subvar[rn, var, var+res[[resn]]]], var];
    qn = Cancel[qn/gn];
    gnj = gn;
    For[i = 1, i <= res[[resn]], i++,
        pn = Expand[pn gnj];
        gnj = Expand[Subvar[gn, var, var-i]]
    ];
    rn = Cancel[rn/gnj]]
];
(* find degree bound *)
qn = Expand[Subvar[qn, var, var+1]];
dp = Exponent[pn, var];
qrpos = Expand[qn+rn]; dqrpos = Exponent[qrpos, var];
qrneg = Expand[qn-rn]; dqrneg = Exponent[qrneg, var];
If[dqrpos <= dqrneg, dk = dp - dqrneg,
    If[dqrneg < dqrpos - 1 || dqrneg == -1, dk = dp - dqrpos + 1,
        (* else *)
        k0 = - Cancel[Coefficient[qrneg, var^dqrneg]/
            Coefficient[qrpos, var^dqrpos]];
        dk = If[IntegerQ[k0], Max[k0, dp - dqrpos + 1],
            dp - dqrpos + 1]
    ]
];
If[dk >= 0, (* solution possible *)
    fn = Sum[f[[ii] var^ii, {ii, 0, dk}];
    eq = Expand[pn - qn fn + rn Subvar[fn, var, var-1]];
    eq = CoefficientList[eq, var];
    eq = Solve[Thread[eq == 0], Array[f, dk+1, 0]];
    If[Length[eq] > 0,
        fn = fn /. eq[[1]];
        fn = fn /. f[0] -> 0; (* summation constant *)
        sn = Cancel[qn/pn fn] ann;
        (* crude test for infinity *)
        s1 = sn /. var -> n1;
        s0 = (sn /. var -> n0) - (ann /. var -> n0);
        Return[s1 - s0],
        (* else fail *)
        Return[Sum[an, {var, n0, n1, 1}]]
    ],
];

```

```

(* else fail *)
    Return[Sum[an, {var, n0, n1, 1}]]
]
End[]

```

Gosper's summation algorithm.

This reads in the symbolic summation package.

`In[1]:= <<Gosper.m`

Here is the result for $\sum_{k=1}^n k^2 (1+k)$.

`In[2]:= Gosper[k^2 (1+k), {k, 1, n}]`

$$\text{Out}[2] = \frac{n (1 + n) (2 + 7 n + 3 n^2)}{12}$$